PEARSON EDEXCEL INTERNATIONAL A LEVEL · CHEMISTRY · WCH12

09

Introduction to Kinetics and Equilibria

Collision theory, Maxwell–Boltzmann, catalysts — and reversible reactions at equilibrium.

Collision theory and rate factors; rates from graphs and tangents; Maxwell–Boltzmann distributions; catalysts and reaction profiles. 9B: reversible reactions and dynamic equilibrium; effects of temperature, pressure and concentration on the position of equilibrium; compromise conditions in the Haber and Contact processes.

UNIT	SECTIONS	STUDY TIME	FLASHCARDS
2 · WCH12	12	~5 h	10

COVERS **12 SPECIFICATION POINTS** — STATEMENTS QUOTED VERBATIM FROM THE SPECIFICATION

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Contents

9.0	Specification Map	SPEC 9.1 - 9.11	8 min
9.1	Collision Theory and Rate Factors	SPEC 9.1	30 min
9.2	Activation Energy, Ea	SPEC 9.2	12 min
9.3	Calculating the Rate of Reaction	SPEC 9.3	24 min
9.4	The Maxwell–Boltzmann Distribution	SPEC 9.4	28 min
9.5	Catalysts: Alternative Routes of Lower Ea	SPEC 9.5	10 min
9.6	Reaction Profiles: Catalysed and Uncatalysed	SPEC 9.6	22 min
9.7	Catalysts in Industry	SPEC 9.7	24 min
9.8	Catalysts on the Maxwell–Boltzmann Curve	SPEC 9.8	12 min
9.9	Reversible Reactions and Dynamic Equilibrium	SPEC 9.9	20 min
9.10	Changing Conditions and the Position of Equilibrium	SPEC 9.10	34 min
9.11	Reversible Reactions in Industry	SPEC 9.11	26 min

HOW TO USE THIS DOCUMENT

Every section opens with the specification statement, worded exactly as Edexcel words it — that is what the examiner marks against. Definitions, equations and exam alerts are boxed; worked examples walk the full method step by step. Read a section, shut the page and recall its bullets from memory, then grind the topic's flashcard deck on the site.

SPEC 9.1 - 9.11 ≈ 8 MIN

Specification Map

THE SPECIFICATION SAYS

Students will be assessed on their ability to:

Every statement the specification assesses in this topic — use it as your revision checklist.

SPEC	ASSESSMENT STATEMENT
9.1	understand, in terms of the collision theory, the effect of changes in concentration, temperature, pressure and surface area on the rate of a chemical reaction
9.2	understand that reactions take place only when collisions have sufficient energy, known as the activation energy
9.3	be able to calculate the rate of a reaction from: i the time taken for a reaction, using rate = 1/time ii the gradient of suitable graph, by drawing a tangent, either for initial rate, or at a time, t
9.4	understand qualitatively, in terms of the Maxwell-Boltzmann distribution of molecular energies, how changes in temperature affect the rate of a reaction
9.5	understand the role of catalysts in providing alternative reaction routes of lower activation energy
9.6	be able to draw the reaction profiles for uncatalysed and catalysed reactions, including the energy level of the intermediate formed with the catalyst
9.7	understand the use of catalysts in industry to make processes more sustainable by using less energy and/or higher atom economy
9.8	be able to interpret the action of a catalyst in terms of a qualitative understanding of the Maxwell-Boltzmann distribution of molecular energies
9.9	know that many reactions are readily reversible and that they can reach a state of dynamic equilibrium in which: i the rate of the forward reaction is equal to the rate of the backward reaction ii the concentrations of the reactants and the products remain constant
9.10	be able to predict and justify the qualitative effects of changes of temperature, pressure and concentration on the position of equilibrium in a homogeneous system
9.11	evaluate data to explain the necessity, for many industrial processes, to reach a compromise between the yield and the rate of reaction

Further suggested practical — experiments to demonstrate the factors that influence the rate of chemical reactions, including the decomposition of hydrogen peroxide, reaction of marble chips with acid, reaction of thiosulfate ions with acid.

Further suggested practical (9B) — demonstrate the effect of a change of temperature, pressure and concentration on a system at equilibrium: chlorine reacting with iodine to form iodine(I) chloride, which then reacts with chlorine to form iodine(III) chloride; and the equilibrium system between nitrogen dioxide (NO_2) and dinitrogen tetroxide (N_2O_4).

The topic is built in two parts: **9A Kinetics** (9.1 – 9.8) and **9B Equilibria** (9.9 – 9.11). Each assessment statement has its own section below, in specification order.

SPEC 9.1 ≈ 30 MIN

Collision Theory and Rate Factors

THE SPECIFICATION SAYS

understand, in terms of the collision theory, the effect of changes in concentration, temperature, pressure and surface area on the rate of a chemical reaction

React only when they collide — hard enough, the right way, and more often when conditions change.

Collision theory

According to the theory, for a reaction to occur —

- The reacting particles must **collide** before a reaction takes place. However, not all collisions will lead to a reaction.
- During collision, the reacting particles must approach each other in the correct orientation.
- The reacting particles must have a certain minimum amount of energy referred to as the activation energy (Ea).

DEFINITION

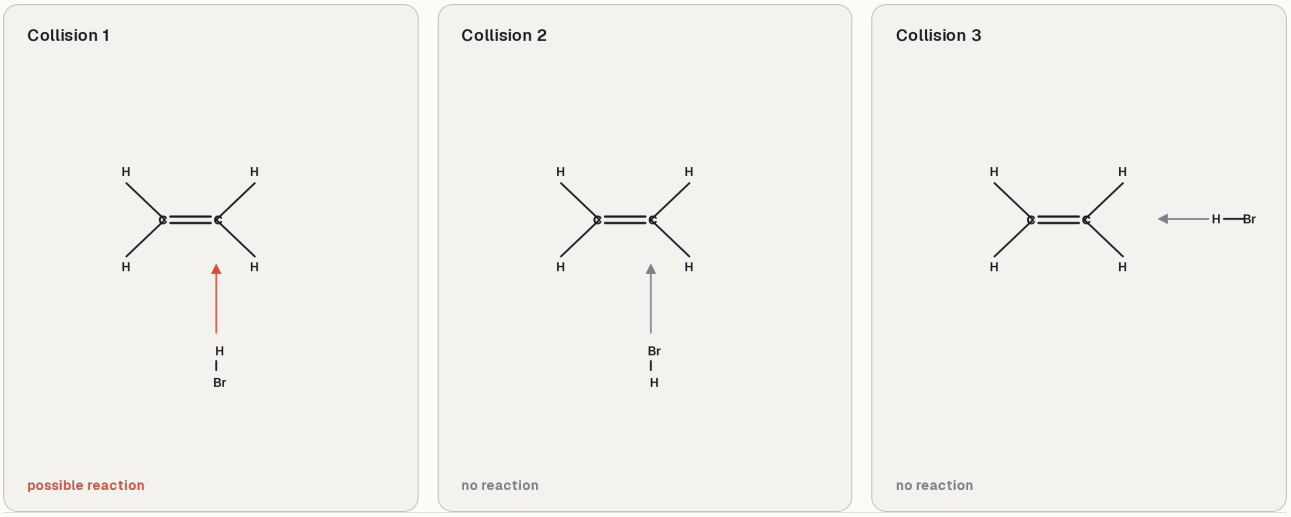
Effective — successful — collisions

A reaction occurs when the reactants collide each other in the correct orientation with energy equal to or greater than activation energy. Such collisions are effective or successful collisions.

Orientation (steric effect)

Collisions will result in reaction only if the molecules collide in the proper geometry.

SPEC 9.1 · COLLISION GEOMETRY **Orientation (steric effect) — only the right geometry reacts** proper geometry



STERIC EFFECT Collisions will result in reaction only if the molecules collide in the proper geometry. Collision 1 brings the H end of H–Br to the C=C in the right orientation — a reaction is possible.

FIG – *Collision 1 — possible reaction · collisions 2 and 3 — no reaction.*

Effect of concentration, pressure and surface area on rate of reaction

- According to collision theory, reactant particles must collide with sufficient energy before they can react.
- Hence rate of reaction can be increased by increasing the frequency of collision with sufficient energy between reacting particles.
- Collisions that result in a reaction are called successful collisions.

The effect of concentration

Increase in the concentration of the reactants in solution increases the rate of the reaction. This is because

- Reactant particles per unit volume increases.
- Frequency of collisions increases.
- So frequency of effective collisions increases.

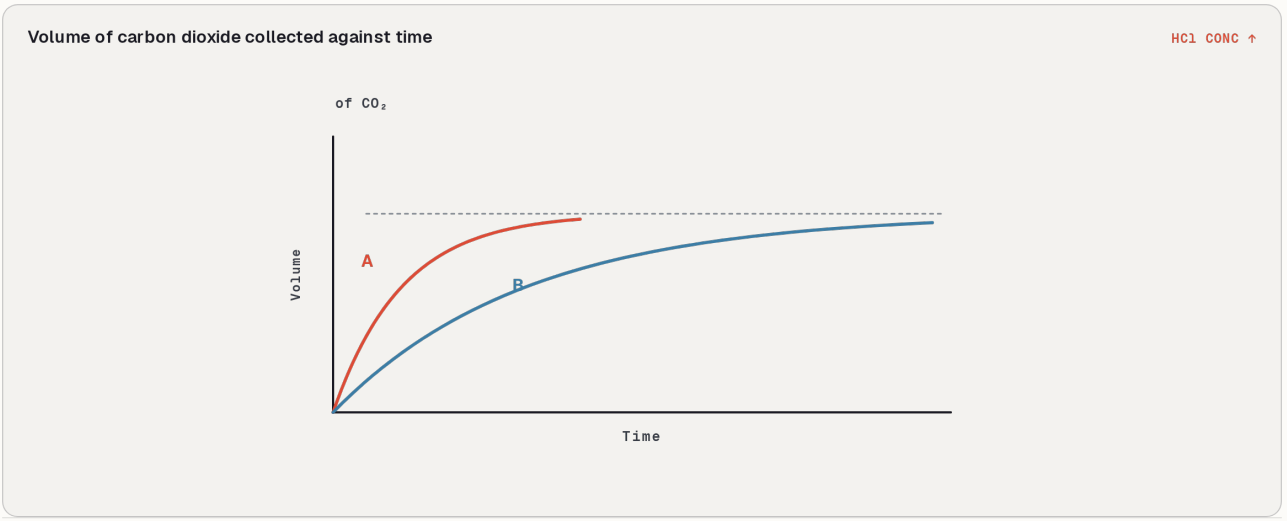
Decreasing the concentration of the reactants in solution decreases the rate of the reaction because

- Reactant particles per unit volume decreases.
- Frequency of collisions decreases.
- So frequency of effective collisions decreases.

Rate of reaction: reactions start off at their fastest then slow as the reactant concentration drops.

eg In the reaction **$CaCO_3(s) + 2HCl \rightarrow CaCl_2 + CO_2(g)$**, the change in volume of carbon dioxide is measured with time when the concentration of HCl acid is increased. The following graph shows the volume of carbon dioxide collected:

SPEC 9.1 · EFFECT OF CONCENTRATION **Higher HCl concentration — steeper curve, faster completion** $CaCO_3(s) + 2HCl \rightarrow CaCl_2 + CO_2(g)$



CURVE A VS B Curve A is steeper therefore rate of reaction is faster than curve B · gradient of curve A is greater than gradient of curve B · curve A levels off quickly compared to curve B indicating faster completion of reaction.

FIG — *Curve A is steeper — rate of reaction is faster than curve B, and the reaction completes faster.*

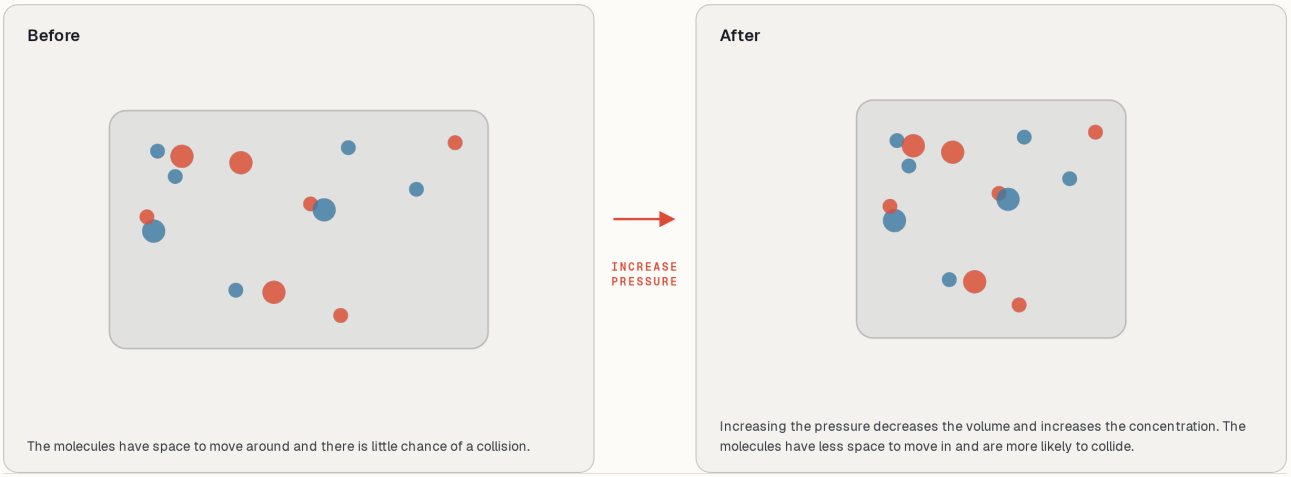
- Curve A is steeper therefore rate of reaction is faster than curve B.
- Gradient of curve A is greater than gradient of curve B.
- Curve A levels off quickly compared to curve B indicating faster completion of reaction.

The effect of pressure (only for gases)

Increasing the pressure of gaseous reacting particles or decreasing the volume of the reaction vessel increases the rate of the reaction. This is because

- Same number of gaseous molecules in smaller space or number of gaseous molecules per unit volume increases.
- Increasing the pressure makes gas particles get closer together.
- This increases the frequency of collisions.
- So frequency of effective collisions increases.

SPEC 9.1 · EFFECT OF PRESSURE (ONLY FOR GASES) **Same molecules, smaller space — more collisions** gases only



PRESSURE ↑ Same number of gaseous molecules in a smaller space, or number of gaseous molecules per unit volume increases · increasing the pressure makes gas particles get closer together · this increases the frequency of collisions · so frequency of effective collisions increases. Changing the pressure has almost no effect on reactions in solid or liquid phase.

FIG — *The molecules have less space to move in and are more likely to collide.*

Changing the pressure has almost no effect on reactions in solid or liquid phase. This is because the volume of solids and liquids changes very little when pressurized. So their particles do not move close together.

The effect of surface area

For heterogeneous reactions increasing surface area leads to more chances of a collision.

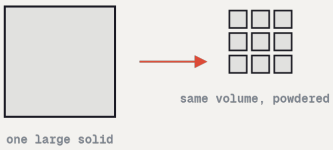
Increasing the surface area of solid reactants increases the rate of the reaction. This is because

- Greater surface area of the solid is exposed for collision.
- Frequency of collisions increases.
- So frequency of effective collisions increases.

Catalysts (in catalytic converters) are used in a finely divided form for this reason.

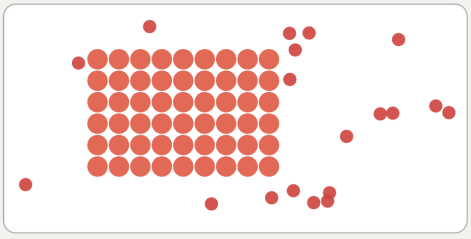
SPEC 9.1 · EFFECT OF SURFACE AREA **Greater surface area — more chances of a collision** heterogeneous reactions

One lump → the same mass as powder



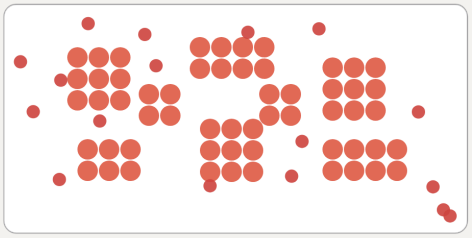
More surface is exposed for reaction.

Solid magnesium + dilute acid



Only collisions between the hydrogen ions and magnesium atoms on the *surface* of magnesium result in a reaction.

Powdered magnesium + dilute acid



If magnesium is powdered, the surface area is increased and hydrogen is given off more quickly.

MORE SURFACE Greater surface area of the solid is exposed for collision · frequency of collisions increases · so frequency of effective collisions increases · catalysts (in catalytic converters) are used in a finely divided form for this reason.

FIG — *More surface is exposed for reaction — only collisions with magnesium atoms on the surface result in a reaction, so powdered magnesium gives off hydrogen more quickly.*

How changes in temperature can affect the rate of a reaction

Increasing the temperature increases the rate of a reaction. This is because

- Average kinetic energy of the particles increases.
- More particles have energy equal to or greater than activation energy.
- Speed of the particles also increases.
- So the frequency of effective collisions increases.

Collisions in solution

- In solution the molecules are much more closely packed together, so there is not much space between them.
- In low to medium concentrations, most of the solution is solvent, so solute particles are surrounded by solvent molecules and are said to be trapped in a **solvent cage**.
- There are a number of solute molecules that get trapped in the same solvent cage. Hence increasing the collision rate between solute particles and if collisions are sufficiently energetic they may react just as in the gas phase.

There are two distinct stages to reaction in solution:

1. Molecules have to come together, by a process of diffusion, in the same solvent cage.
2. They have to react.

For simple reactions involving two species, the chance of them meeting each other in solution is proportional to their concentrations, same as in gas phase.

SPEC 9.2 ≈ 12 MIN

Activation Energy, Ea

THE SPECIFICATION SAYS

understand that reactions take place only when collisions have sufficient energy, known as the activation energy

Collisions need enough energy to break bonds — otherwise they simply bounce apart.

DEFINITION

Activation energy, Ea

Activation energy is the minimum energy that colliding particles must possess for a reaction to occur.

- If particles collide with less energy than activation energy then they simply bounce apart and no reaction occurs.
- Only those collisions that have energies equal to or greater than activation energy result in a reaction.
- Chemical reaction results in the breaking of some bonds (needing energy) and making new bonds (releasing energy). Activation energy is involved in breaking some of the original bonds.
- If collisions are relatively gentle, there is not enough energy available to break the bonds and so particles do not react.

SPEC 9.3 ≈ 24 MIN

Calculating the Rate of Reaction

THE SPECIFICATION SAYS

be able to calculate the rate of a reaction from: i the time taken for a reaction, using rate = 1/time ii the gradient of suitable graph, by drawing a tangent, either for initial rate, or at a time, t

Rate from concentration–time graphs: tangents, gradients, initial rate.

Rate of reaction

Chemical kinetics is concerned with the dynamics of chemical reactions such as the way reactions take place and the rate (speed) of the process.

Rate of reaction can be defined as

- The **decrease** in the concentration of any one of the reactants per unit time
- **OR** the **increase** in the concentration of any one of the product per unit time

Rate of reaction = change in concentration / time for change to happen

In a chemical reaction, the concentration of the reactants decreases or the concentration of the products increases.

A graph can be drawn by plotting the concentration of the reactants or products against time; the slope of the graph measures the rate of the reaction.

To measure rate of a reaction, we need to find out:

- How fast one of the reactants is used up
- How fast one of the products is formed

SPEC 9.3 · RATE FROM GRAPHS **Rate from concentration–time graphs** tangent → gradient → rate



GRADIENT = Y/X The slope of the graph measures the rate of the reaction. Y = change of concentration (mol dm⁻³) · X = change in time (seconds) · unit of rate = mol dm⁻³ s⁻¹ · rate is determined by drawing a tangent at a point and measuring its gradient.

FIG — *fig A — with respect to reactants: tangent at point A, gradient zero when the reaction is complete · fig B — with respect to products: initial rate, rate at time t, rate at end = zero.*

Rate of reaction with respect to reactants

- The gradient decreases as the rate decreases and becomes zero when all reactants are used up.
- Rate of reaction is determined by drawing a tangent at a point (eg point A) and measuring its gradient.

Gradient = y / x

Y = change of concentration (mol dm⁻³) · X = change in time (seconds)

Unit of rate = mol dm^{-3} s^{-1}

Rate of reaction with respect to products

- Rate is also measured by the change in concentration of products over a period of time.

- The slope at the start of the reaction will give the **initial rate**.
- The slope gets less (showing the rate is slowing down) as the reaction proceeds.

SPEC 9.4 ≈ 28 MIN

The Maxwell–Boltzmann Distribution

> THE SPECIFICATION SAYS
>
> **understand qualitatively, in terms of the Maxwell-Boltzmann distribution of molecular energies, how changes in temperature affect the rate of a reaction**

One curve explains why a small temperature rise gives a large rate increase.

Maxwell-Boltzmann distribution curves

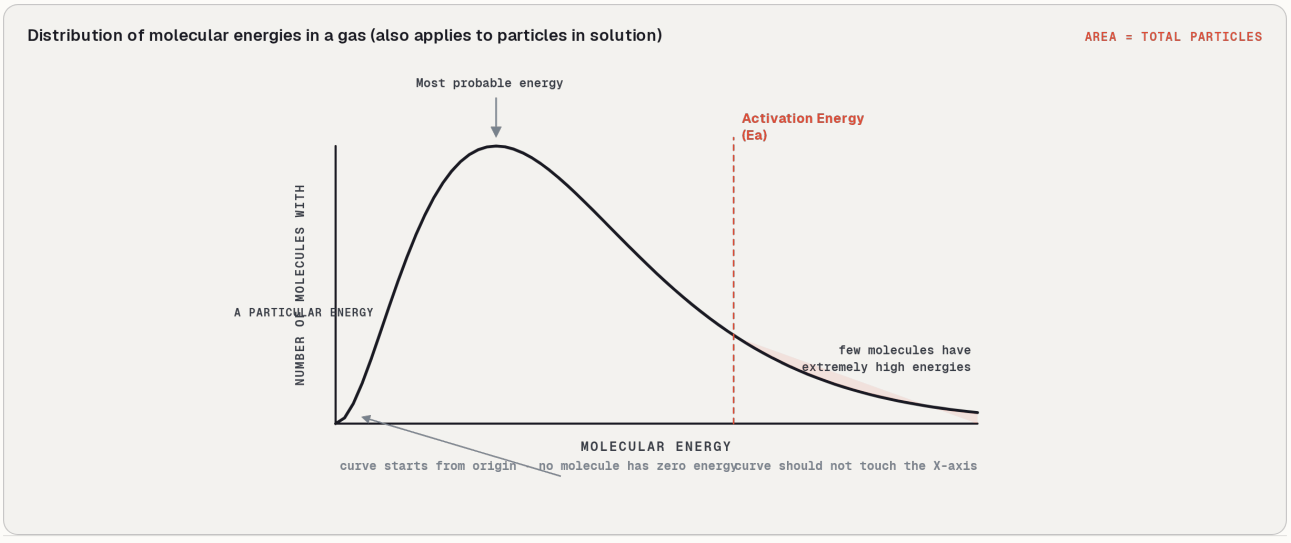
- The molecules in a sample of gas have a wide range of energies. Due to the number of collisions taking place, there is a spread of molecular velocities and energies.

Experiments indicated that ...

- No particles have zero velocity (all the particles have some amount of energy).
- Some have very low and some have very high velocities.
- Most have intermediate velocities.
- Area under the curve represents the total number of particles in the system.

This curve shows the distribution of energies of the molecules in a gas (but can also be applied for particles in a solution).

SPEC 9.4 · MAXWELL-BOLTZMANN DISTRIBUTION CURVES **Molecules have a wide range of energies** spread of velocities



DISTRIBUTION The area under the curve beyond Ea is proportional to the number of particles with energy greater than activation energy · if the activation energy is high, only about 1 in 10^{12} molecules possesses this energy at room temperature, so the reaction does not take place — the system is said to be kinetically stable.

FIG — *Curve starts from origin · no molecule has zero energy · few molecules have extremely high energies · the curve should not touch the X-axis.*

- **Ea** corresponds to the activation energy.
- The area under the curve beyond this Ea value is proportional to the number of particles with energy greater than activation energy.

If the **activation energy is high**, the number of colliding particles that have this energy at room temperature is extremely small (only about 1 in 10^{12} molecules possesses this energy), so the reaction does not take place. The system is said to be **kinetically stable**.

Such reactions occur only if the temperature is increased.

Effect of an increase in temperature on the Maxwell-Boltzmann curve

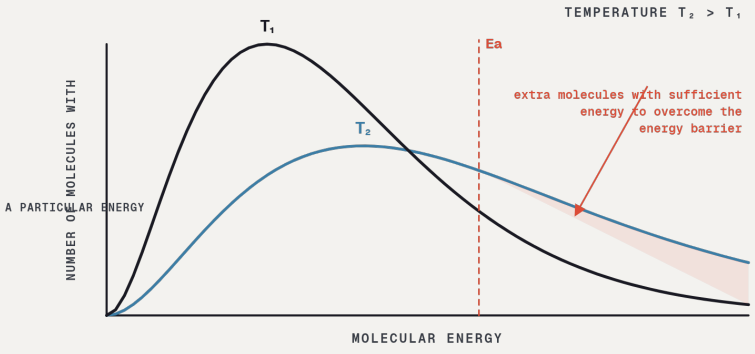
Increasing the temperature alters the distribution:

- Curve shifts to the **right** — higher energies/velocities.
- Curve gets **broader and flatter** due to the greater spread of values.
- Area under the curve stays constant — corresponds to the total no. of particles.

SPEC 9.4 · TEMPERATURE AND THE CURVE **Higher temperature — more particles beyond Ea** $T_2 > T_1$

Effect of an increase in temperature on the Maxwell-Boltzmann curve

AREA STAYS CONSTANT



$T_2 > T_1$ Curve shifts to the RIGHT, higher energies/velocities · curve gets broader and flatter due to the greater spread of values · area under the curve stays constant — corresponds to the total no. of particles · neither curve is symmetrical · both curves start at the origin and finish by approaching the X axis and do not touch the X axis · the area under both curves is same.

FIG — *TEMPERATURE $T_2 > T_1$ — the shaded area is the extra molecules with sufficient energy to overcome the energy barrier.*

- Increasing the temperature gives more particles with energy greater than activation energy (Ea).
- More reactants can overcome the energy barrier associated and form products.
- Therefore, more effective collisions can take place, increasing the rate of reaction.
- A small rise in temperature can lead to a large increase in rate.

Points to note about the Maxwell-Boltzmann graph

- Neither curve is symmetrical.
- Both curves start at the origin and finish by approaching the X axis and do not touch the X axis.
- The area under both curves is the same.

- The peak of the T_2 distribution is to the right and lower than the peak of the T_1 distribution.

SPEC 9.5 ≈ 10 MIN

Catalysts: Alternative Routes of Lower Ea

> THE SPECIFICATION SAYS
>
> **understand the role of catalysts in providing alternative reaction routes of lower activation energy**

Catalysts alter the rate without undergoing any permanent chemical change.

The effect of catalysts

- A catalyst alters rate of reaction by providing an alternative path with lower activation energy — **but** they do not undergo any permanent chemical change.
- Catalysts are specific to reactions and they cause the reaction rate to increase.

SPEC 9.6 ≈ 22 MIN

Reaction Profiles: Catalysed and Uncatalysed

> THE SPECIFICATION SAYS
>
> **be able to draw the reaction profiles for uncatalysed and catalysed reactions, including the energy level of the intermediate formed with the catalyst**

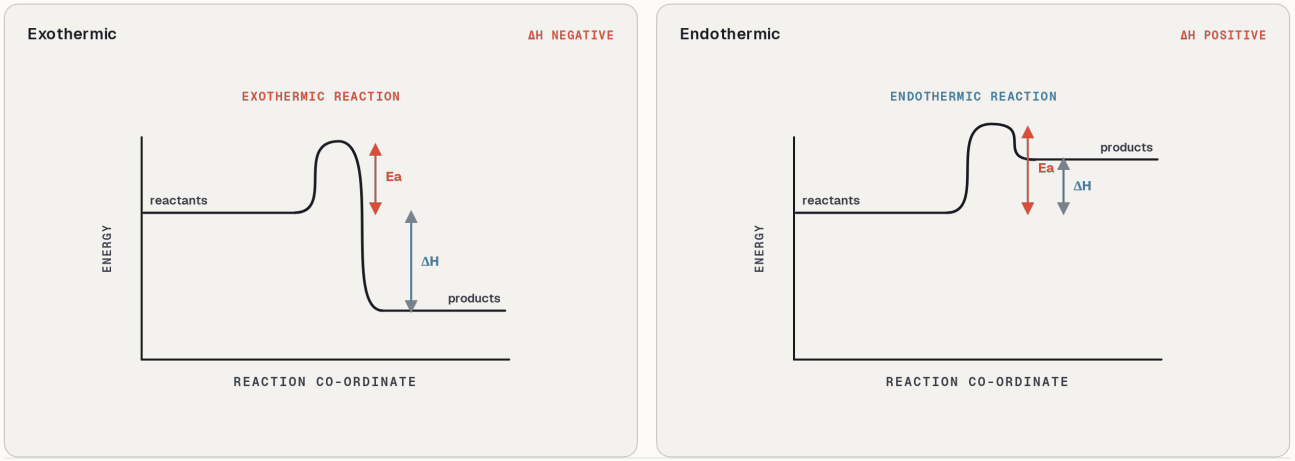
Profiles with Ea, ΔHr — and the intermediate formed with the catalyst.

Reaction profile diagrams

A reaction profile diagram shows the energy levels of the reactants and the products of the reaction and of the transition state that the reactants go through. The activation energy (Ea) and the enthalpy change (ΔHr) are also shown on the diagram.

The energy changes during the course of an exothermic and endothermic reaction are shown in the reaction profile diagram.

SPEC 9.6 · REACTION PROFILE DIAGRAMS **Energy levels from reactants to products** Ea and ΔHr shown

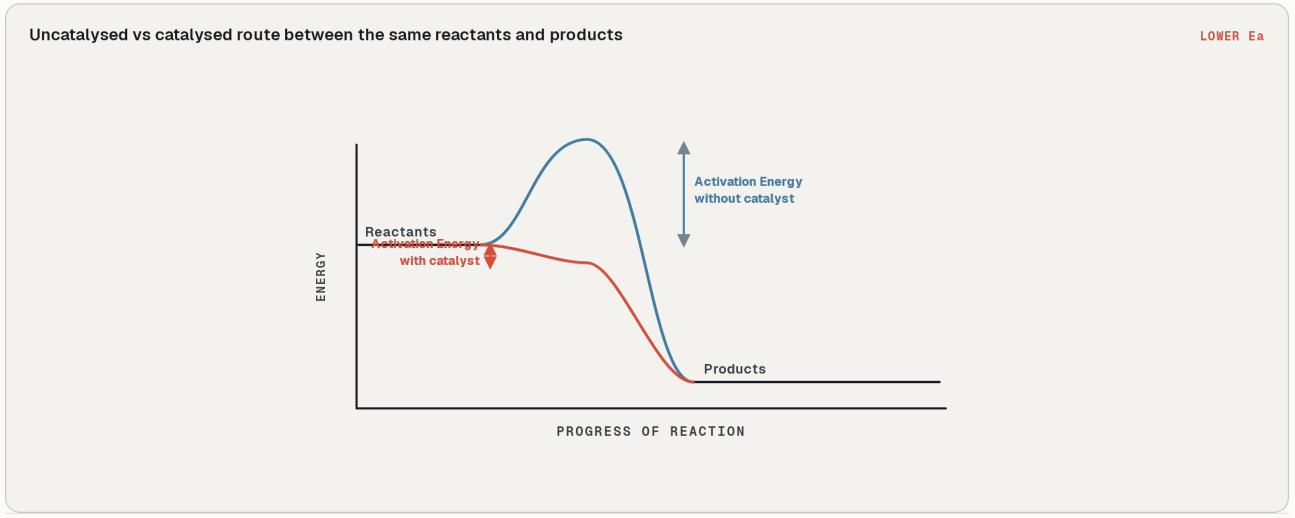


EXO / ENDO Reaction profile diagram shows the energy levels of the reactants and the products of the reaction and of the transition state that the reactants go through · the activation energy (Ea) and the enthalpy change (ΔHr) are also shown on the diagram.

FIG — *EXOTHERMIC REACTION — products lower than reactants · ENDOTHERMIC REACTION — products higher than reactants · Ea and ΔH shown on both.*

A catalyzed reaction takes place using a different route from the uncatalyzed reaction.

SPEC 9.6 · ALTERNATIVE ROUTE **A catalysed reaction takes a different route** same ΔH, lower Ea



LOWER EA A catalyzed reaction takes place using a different route from the uncatalyzed reaction — the catalysed hump is lower, so more collisions succeed.

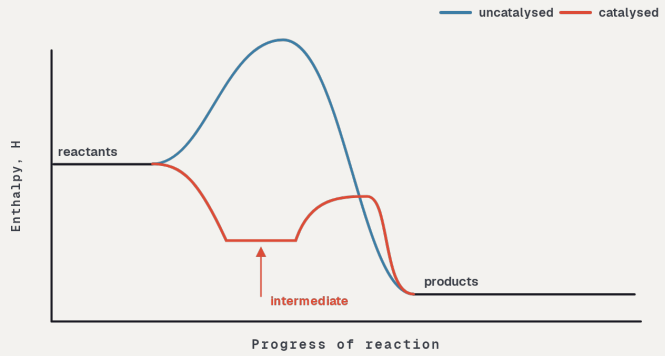
FIG — *Activation energy without catalyst is higher; activation energy with catalyst is lower.*

If an intermediate is formed between the reactant and the catalyst then the reaction takes place in at least two steps as shown below.

Example: the reaction between peroxydisulphate ions and iodide ions, catalyzed by Fe^{2+} ions.

SPEC 9.6 · INTERMEDIATE FORMED WITH THE CATALYST **At least two steps when an intermediate forms** — Fe^{2+}-catalysed $S_2O_8^{2-}$ + I^-

Enthalpy profile diagram for a catalyzed reaction involving the formation of an intermediate TWO STEPS



TWO STEPS If an intermediate is formed between the reactant and the catalyst then the reaction takes place in at least two steps — example: the reaction between peroxydisulphate ions and iodide ions, catalyzed by Fe^{2+} ions.

FIG — *Enthalpy profile diagram for a catalyzed reaction involving the formation of an intermediate.*

SPEC 9.7 ≈ 24 MIN

Catalysts in Industry

THE SPECIFICATION SAYS

understand the use of catalysts in industry to make processes more sustainable by using less energy and/or higher atom economy

Homogeneous and heterogeneous catalysts — less energy, higher atom economy.

Catalysts in industry

Catalysts have been increasingly used in the chemical industry. The development of catalysts was mostly for economic reasons, but now development of catalysts has been for political and environmental reasons as well.

Two major economic advantages of the use of catalysts are:

1. Increase the rate of chemical reaction — more of the desired product can be made in a given period of time.
2. Reactions can take place at lower temperatures, resulting in a decrease in the energy costs to the manufacturer.

Homogeneous catalyst

DEFINITION

Homogeneous catalyst

A homogeneous catalyst is one that is in the same phase (physical state) as the reactants.

Example: Fe^{2+} ion catalyst in oxidation of iodide ions I^- by persulphate ions $S_2O_8^{2-}$.

$$S_2O_8^{2-}(aq) + 2I^-(aq) \xrightarrow{Fe^{2+}(aq)} 2SO_4^{2-}(aq) + I_2(s)$$

Both the reactants and the iron(II) catalyst are in the aqueous phase.

Heterogeneous catalyst

Most catalysts used in industry are heterogeneous catalysts.

- A heterogeneous catalyst is one that is in a different phase from the reactants.
- Solids are commonly used as heterogeneous catalysts for reactions involving gases.

Example: many reactions between gases are catalyzed using a solid catalyst.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

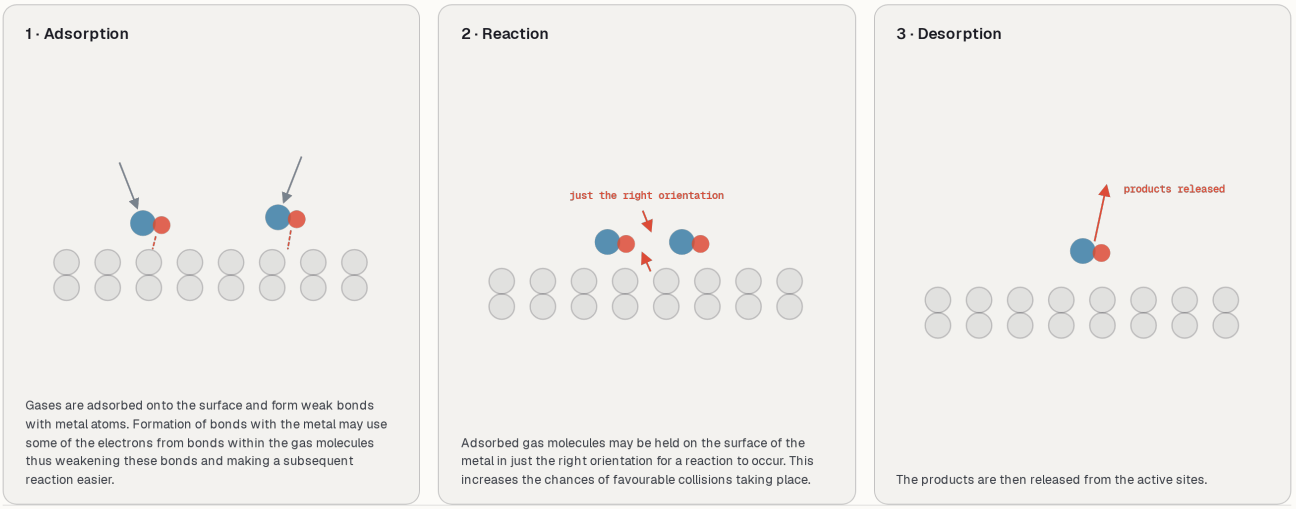
Iron powder as a catalyst — the reactants are in the gaseous state, but the catalyst is in the solid state.

How heterogeneous catalysts work

Heterogeneous catalysis takes place at active sites on the surface of a solid (e.g. a metal). Gases are adsorbed onto the surface and form weak bonds with metal atoms. This form of catalysis is thought to work in three stages as follows:

STAGE	WHAT HAPPENS
1. Adsorption	Formation of bonds with the metal may use some of the electrons from bonds within the gas molecules thus weakening these bonds and making a subsequent reaction easier.
2. Reaction	Adsorbed gas molecules may be held on the surface of the metal in just the right orientation for a reaction to occur. This increases the chances of favourable collisions taking place.
3. Desorption	The products are then released from the active sites.

SPEC 9.7 · HOW HETEROGENEOUS CATALYSTS WORK **Active sites on the surface of a solid** three stages



ACTIVE SITES Heterogeneous catalysis takes place at active sites on the surface of a solid (e.g. a metal) — in the example, H_2 and C_2H_4 adsorb on the metal surface, the H–H bond breaks and H atoms migrate along the surface, and ethane (C_2H_6) desorbs.

FIG — *In the example, H_2 and C_2H_4 adsorb on the metal surface, the H–H bond breaks and H atoms migrate along the surface, and ethane (C_2H_6) desorbs.*

Catalysts and atom economy in industry

- In the manufacture of chemicals, waste is produced when products are formed.
- Use of catalysts can lead to reactions with little or no co-product (waste). It may also reduce side reactions producing unwanted products, making separation easier.
- Example: phenol can be manufactured by the oxidation of dinitrogen monoxide (N_2O) using a zeolite catalyst, giving nitrogen as the only by-product.
- Manufacture of phenol using benzene via cumene has a very low atom economy.

Advantages of using catalysts

- Fewer steps are involved in the manufacture, hence the energy costs are likely to be less.
- The percentage yield of each step may be higher (and fewer steps).
- Less product is lost in the separation after each step, or separation may not be required until the end.
- The whole process may be quicker so that more product is obtained in a given time.

SPEC 9.8 ≈ 12 MIN

Catalysts on the Maxwell–Boltzmann Curve

THE SPECIFICATION SAYS

be able to interpret the action of a catalyst in terms of a qualitative understanding of the Maxwell-Boltzmann distribution of molecular energies

Lower the Ea line and more particles can react.

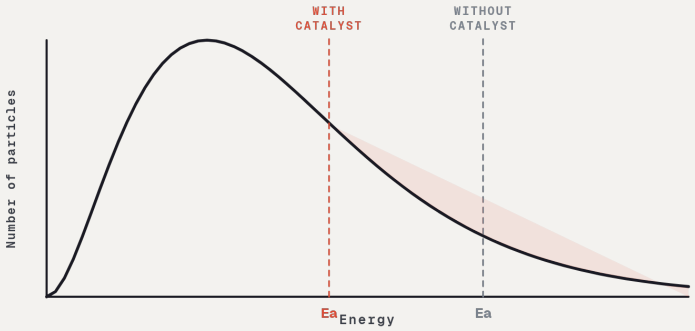
Maxwell-Boltzmann distribution curve for catalysed reactions

- Catalysts provide an alternative pathway with lower activation energy.
- Since activation energy is reduced, more particles have sufficient energy for effective collisions.
- Therefore, the number of effective collisions increases, thus increasing the rate of reaction.
- Using catalysts avoids the need to supply extra heat — it is safer and cheaper.
- Catalysts remain chemically unchanged at the end of the reaction.

SPEC 9.8 · CATALYSTS ON THE DISTRIBUTION **Lower Ea — more particles can react** shaded area grows

Maxwell-Boltzmann distribution curve

LOWER Ea LINE



LOWER EA LINE Catalysts provide an alternative pathway with lower activation energy · since activation energy is reduced, more particles have sufficient energy for effective collisions · therefore, number of effective collisions increases, thus increases the rate of reaction · using catalysts avoids the need to supply extra heat — it is safer and cheaper · catalysts remain chemically unchanged at the end of the reaction.

FIG – *The Ea line with a catalyst is lower — the shaded area of particles with sufficient energy is larger, so the rate increases.*

SPEC 9.9 ≈ 20 MIN

Reversible Reactions and Dynamic Equilibrium

> THE SPECIFICATION SAYS
>
> **know that many reactions are readily reversible and that they can reach a state of dynamic equilibrium in which: i the rate of the forward reaction is equal to the rate of the backward reaction ii the concentrations of the reactants and the products remain constant**

Reversible reactions, the ⇌ symbol — and how forward and backward rates become equal in a closed container.

Irreversible and reversible reactions

When a mixture of hydrogen and oxygen in a 2:1 ratio is ignited with a spark, water is produced and no uncombined hydrogen and oxygen is left. This is an example of a complete or irreversible reaction.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

But many reactions do not go to completion. For example, a mixture of hydrogen and iodine in a 1:1 molar ratio is heated to 300 °C in a closed vessel, only about 90% of hydrogen and iodine reacts. This is an example of a **reversible reaction**.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

- A reversible reaction is one that goes in both directions at the same temperature.
- The symbol ⇌ is used to indicate reactions that are reversible.
- Many reactions go to completion if they are performed in an open container.

Example: $CaCO_3(s)$ would completely decompose into $CaO(s)$ and $CO_2(g)$ on heating at 900 °C in an open container according to the equation;

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

- However, if the above reaction was performed in a closed container, only a small portion of $CaCO_3$ decomposes, no matter how long the temperature was maintained at 900 °C.
- The reason for this is that the products CaO and CO_2 can also react together at this temperature to form the reactants $CaCO_3$. The reaction is reversible.

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

In a reversible reaction, there are actually two reactions: one from the reactant side to the product side (left-hand side to right-hand side), known as the **forward reaction**, and the other one from the right-hand side to the left-hand side, the **reverse (backward) reaction**.

How is equilibrium established?

- In a reversible reaction, as the reaction continues the concentration of the reactants decreases and the rate of the forward reaction also decreases.
- The following reaction proceeds in the forward direction because no products are yet present.

$$A + B \rightarrow$$

As the reaction proceeds, the products formed react together to re-form the reactants. The rate of forward reaction decreases with time.

$$A + B \rightleftharpoons C + D$$

Since the concentration of the product is initially low, the rate of the backward reaction is also slow. As the concentration of the product increases the rate of the backward reaction increases.

SPEC 9.9 · HOW IS EQUILIBRIUM ESTABLISHED? **One rate falls, the other rises — until they are equal** A + B ⇌ C + D



FORWARD VS BACKWARD As the reaction continues the concentration of the reactants decreases and the rate of the forward reaction also decreases · the rate of the backward reaction is initially slow — as the concentration of the product increases it increases too · eventually the two rates become equal: a dynamic equilibrium has been reached.

FIG — *Rate of A + B reaction falls with time · rate of C + D reaction increases with time.*

Eventually the rates of the forward reaction and backward reaction become equal. At this point a dynamic equilibrium has been reached.

Dynamic equilibrium

Two conditions must be met for dynamic equilibrium to be established;

1. The reaction must be reversible.
2. The reaction mixture must be in a closed container.

SPEC 9.9 · DYNAMIC EQUILIBRIUM **Forward and backward rates become equal** closed system

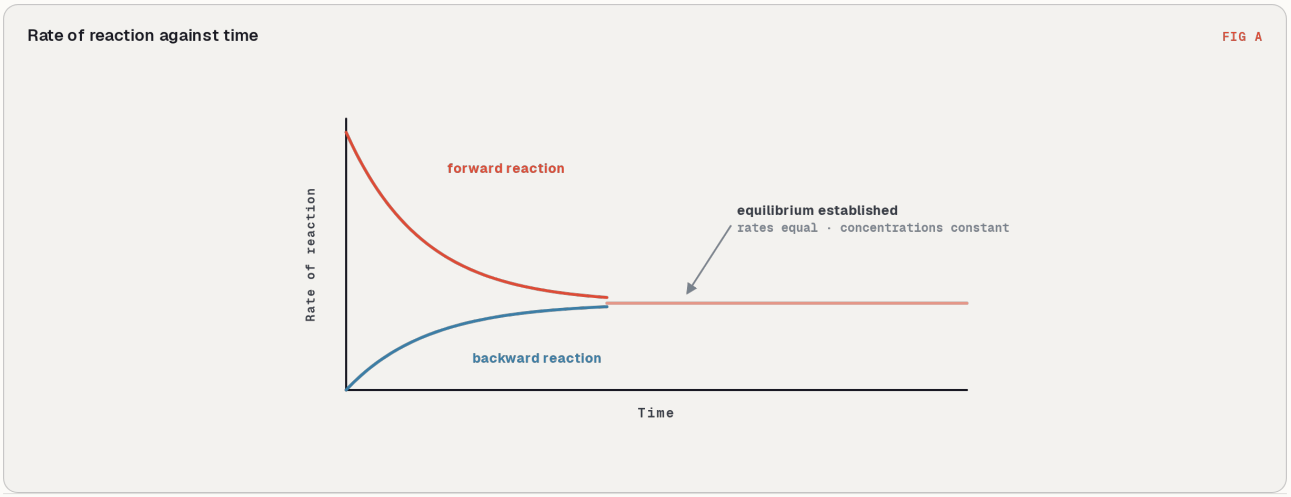


FIG A Two conditions must be met: the reaction must be reversible, and the reaction mixture must be in a closed container · the reactions continue to occur in both directions · the concentrations of the reactants and products remain constant · rate of the forward reaction is equal to the rate of the backward reaction.

FIG – *fig A — Dynamic equilibrium.*

Three important features of a system in dynamic equilibrium:

- The reactions continue to occur in both directions.
- The concentration of the reactants and products remain constant.
- Rate of the forward reaction is equal to the rate of the backward reaction.

SPEC 9.10 ≈ 34 MIN

Changing Conditions and the Position of Equilibrium

> THE SPECIFICATION SAYS
>
> **be able to predict and justify the qualitative effects of changes of temperature, pressure and concentration on the position of equilibrium in a homogeneous system**

Concentration, pressure, temperature and catalysts — predicting and justifying how the position of equilibrium shifts.

Types of equilibria

Homogeneous equilibrium: all reactants and products are in the same physical state (e.g. all gases or all liquids).

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

All reactants and products in gaseous state.

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

All reactants and products in liquid state.

Heterogeneous equilibrium: all reactants and products are in different physical states.

$$H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$$

$$Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$$

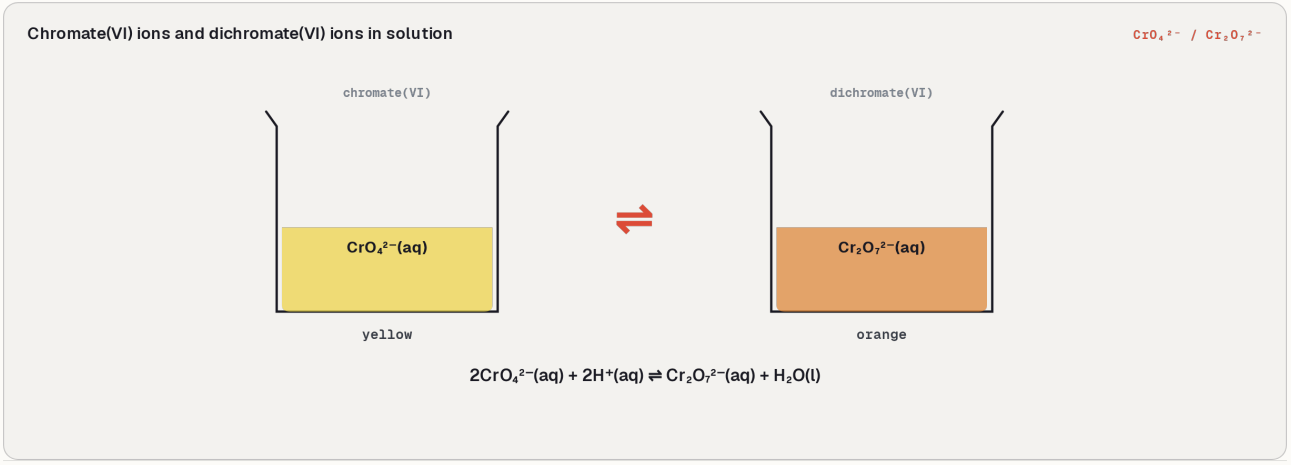
All reactants and products in solid and aqueous state.

Changing the composition of an equilibrium mixture

- When a reaction mixture reaches a position of equilibrium, the concentration of each component will not alter as long as the conditions remain constant.
- However, if a condition is changed by adding more of or removing one of the reactants or products, then the composition of the components may change.
- This is referred to as "changing the position of equilibrium" and we can refer to the position being moved to the right, to the left or unchanged.

For example: reaction between chromate(VI), CrO_4^{2-} ions with acid turns orange. This is because of an increase of dichromate(VI) ions, $Cr_2O_7^{2-}$.

SPEC 9.10 · CHANGING THE POSITION OF EQUILIBRIUM **Chromate(VI) — dichromate(VI) equilibrium** concentration



YELLOW ⇌ ORANGE If acid (H^+) is added to the system, equilibrium moves to the right and the solution becomes orange · if alkali (OH^-) is added to the system, equilibrium moves to the left and the solution becomes yellow.

FIG — *CrO_4^{2-} — yellow · $Cr_2O_7^{2-}$ — orange.*

- If acid (H^+) is added to the system, equilibrium moves to the right and solution becomes orange.
- If alkali (OH^-) is added to the system, equilibrium moves to the left and solution becomes yellow.

Four factors that may affect the position of equilibrium

1. Concentration of component.
2. Pressure of the system.
3. Temperature of the system.
4. Addition of a catalyst.

Effect of changes in concentration on the position of equilibrium

- Increasing concentration of reactants drives the forward reaction. Therefore, the equilibrium shifts to the right.
- Increasing concentration of products drives the backward reaction. Therefore, the equilibrium shifts to the left.
- As the concentration of the products increases, the rate of backward reaction increases and a new equilibrium is established.
- The position of equilibrium has moved to the right, with slightly more products than at the original position of equilibrium.

Summary

CONCENTRATION CHANGE	EFFECT ON EQUILIBRIUM
Reactants increase	Moves to right (rate of forward reaction increases to decrease the concentration of reactants)
Reactants decrease	Moves to left (rate of backward reaction increases to increase the concentration of the reactants)
Products increase	Moves to left (rate of backward reaction increases to decrease the concentration of the products)
Products decrease	Moves to right (rate of forward reaction increases to increase the concentration of the products)

Effect of changes of pressure on the position of equilibrium

- Effect of pressure only applies to equilibrium reactions involving gases.
- If there is no change in the number of moles of gas in a gaseous reaction, changes of pressure have no effect on the composition of equilibrium mixture.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

2 moles ⇌ 2 moles

Applies only to systems where the total number of moles of gas on the reactant side is different from that on the product side. This is because the number of moles of a gas is proportional to the pressure.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

3 moles ⇌ 2 moles

- If the pressure of an equilibrium mixture is increased at constant temperature, the position of the equilibrium will shift towards the side with fewer number of gas molecules, so that the total pressure is decreased.

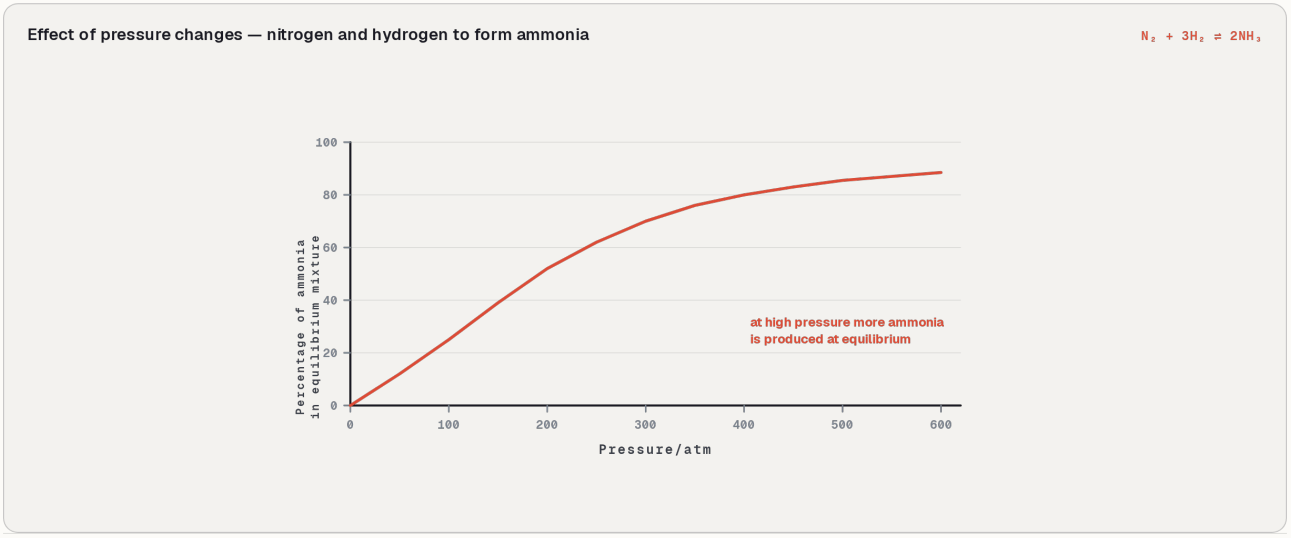
- The pressure of an equilibrium system can be increased by reducing the volume, or adding more gas into the same volume.
- If the pressure of an equilibrium mixture is decreased at constant temperature, the position of equilibrium will shift towards the side with greater number of gas molecules, so that the total pressure is increased.
- Position of equilibrium moves to restore equilibrium.

Summary

PRESSURE CHANGE	EFFECT ON EQUILIBRIUM
INCREASE	Moves to side with fewer gaseous molecules
DECREASE	Moves to side with more gaseous molecules

Consider the reaction between nitrogen and hydrogen to form ammonia. The following graph shows the effect of pressure changes using different amounts of nitrogen and hydrogen to form ammonia:

SPEC 9.10 · EFFECT OF PRESSURE **Percentage of ammonia in the equilibrium mixture against pressure** 4 moles ⇌ 2 moles



PRESSURE ↑ At high pressure more ammonia is produced at equilibrium · high pressure shifts equilibrium to the side having fewer moles of gas.

FIG — *At high pressure more ammonia is produced at equilibrium — high pressure shifts equilibrium to the side having fewer moles of gas.*

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

4 moles ⇌ 2 moles

Effect of changes of temperature on the position of equilibrium

$$\text{Reactants} \rightleftharpoons \text{Products}$$

Forward reaction: ΔH = −ve · Backward reaction: ΔH = +ve

- The direction of change in equilibrium position depends upon whether the reaction is exothermic or endothermic.
- If the temperature of an equilibrium system is increased, the position of equilibrium shifts towards the endothermic direction.
- If the temperature of an equilibrium system is decreased, the position of equilibrium shifts towards the exothermic direction.
- Position of equilibrium moves to establish a new equilibrium.

Summary

TYPE OF FORWARD REACTION	ΔH	INCREASE TEMPERATURE	DECREASE TEMPERATURE
Exothermic	−	Moves to LEFT	Moves to RIGHT
Endothermic	+	Moves to RIGHT	Moves to LEFT

Effect of the addition of a catalyst

- A catalyst works by providing an alternative route that has lower activation energy than that of uncatalysed reaction.
- Therefore rate of a reaction with a catalyst is much faster than the rate without the catalyst.
- A catalyst has **NO EFFECT** on the position of equilibrium of a reversible reaction.

> KEY POINT
>
> **Catalyst**
>
> Catalyst speeds up the forward and backward reactions equally, so equilibrium is reached sooner than without catalyst.

Limitations of making qualitative predictions

- The qualitative predictions made about the effect of concentration, pressure and temperature on the position of equilibrium are just predictions. The arguments are not explanations of why changes sometimes occur.
- There are occasions when it is impossible to predict the direction of change or indeed when predictions turn out to be incorrect.

Example: the equilibrium mixture of $NO_2(g)$ (brown) and $N_2O_4(g)$ (colourless) in a closed container is placed into a beaker of hot water at room temperature — both the temperature and the pressure of the gaseous mixture will rise.

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \quad \Delta H = -57.2 \text{ kJ mol}^{-1}$$

brown ⇌ colourless

- Forward reaction is exothermic so we would predict an increase in temperature would shift equilibrium to the left. But since there are fewer moles on the right-hand side, we would predict that an increase in pressure would shift the equilibrium to the right.
- We do not know which effect is greater, so we cannot make a prediction of which way equilibrium will shift.
- In practice the mixture becomes darker in colour, so the temperature effect must be greater than the pressure effect. The equilibrium must have shifted to the left to produce more $NO_2(g)$.

SPEC 9.11 ≈ 26 MIN

Reversible Reactions in Industry

> THE SPECIFICATION SAYS
>
> **evaluate data to explain the necessity, for many industrial processes, to reach a compromise between the yield and the rate of reaction**

Yield versus rate — compromise conditions in the Haber process and the Contact process.

Applying the principles of reaction rates and reversibility to industrial processes

- The principles of reaction rates and reversibility are important in the design and conditions of many industrial processes.
- To maximize profits, it is important to convert reactants to products (1) as quickly as possible and (2) as completely as possible.
- The first one relates to the kinetics (rate of reaction) and the second is related to reversibility.
- The solution to the problem is a careful choice of reaction conditions.
- The best conditions in an industry are known as **compromise conditions**.

The Haber process

Ammonia is manufactured in industry by direct synthesis from nitrogen and hydrogen.

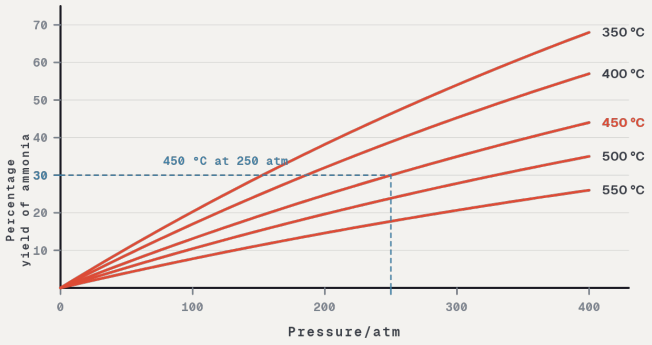
$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92.4 \text{ kJ mol}^{-1}$$

- If the reaction mixture were to reach equilibrium, maximum yield is obtained by low temperature (forward reaction is exothermic) and high pressure (4 mol of gas on the left; 2 mol of gas on the right).
- However, the reaction mixture does not reach equilibrium in the reaction chamber.

SPEC 9.11 · THE HABER PROCESS $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ low T · high P = max yield

The graph shows the equilibrium yield of ammonia at different temperatures and pressures

$\Delta H = -92.4$ kJ mol^{-1}



350 - 550 °C If the reaction mixture were to reach equilibrium, maximum yield is obtained by low temperature (forward reaction is exothermic) and high pressure · conditions used: temperature 450 °C, pressure 250 atm — a compromise between yield and rate.

FIG — *The graph shows the equilibrium yield of ammonia at different temperatures and pressures.*

Why are conditions of 450 °C and 250 atm used?

- Reactions between nitrogen and hydrogen are very slow at room temperature mainly due to the very strong nitrogen to nitrogen triple bond [E(N≡N) = 945 kJ mol^{-1}] producing a high activation energy for the reaction.
- Even at high temperature the reaction is slow in the absence of a suitable catalyst.
- Many metals including tungsten and platinum catalyse this reaction. However, they are very expensive, so iron is used in the industrial process.
- Catalysts require high temperatures to function efficiently. But high temperatures are not economical because of the extra energy costs involved.
- Increasing temperature might also result in a decrease in yield as the forward reaction is exothermic, leading to an increase in the rate of backward reaction.
- In reality these industrial processes are not in equilibrium because the products are removed from the system as they are formed.
- Hence a compromised temperature of 450 °C is used.
- The higher the pressure the larger is the yield, but the larger costs for compressing gases makes the process less economical.
- Hence a compromise is reached between yield and cost — a pressure of 250 atm is used.

Conditions used in the Haber process are;

TEMPERATURE	PRESSURE
450 °C	250 atm

The Contact process

Sulphuric acid is manufactured by the Contact process.

$$SO_2(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \qquad \Delta H = -96\ kJ\ mol^{-1}$$

- Since forward reaction is exothermic, low temperature would result in a high yield of $SO_3(g)$.
- The catalyst vanadium(V) oxide requires high temperature to be effective. So, a moderately high temperature of 450 °C and 1 atm pressure is used.
- At 1 atm pressure the yield is already very high at 97%. Higher pressures would increase the yield as there are fewer molecules on the right hand side.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

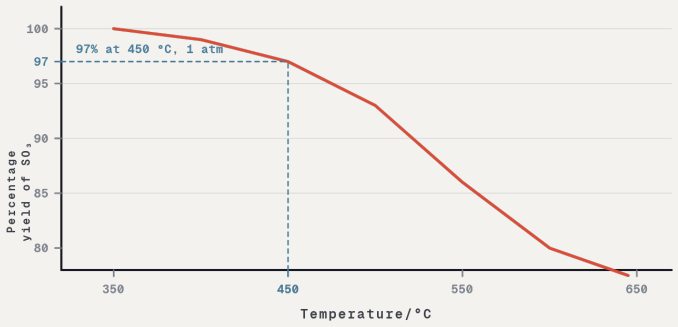
3 moles ⇌ 2 moles

Pressures above 2 atm are unnecessary since the yield is already very high.

SPEC 9.11 · THE CONTACT PROCESS $SO_2(g) + \tfrac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$ 450 °C · 1 atm

Graph showing the percentage yield of sulphur trioxide against temperature, at a pressure of 1 atm

ΔH = −96 kJ mol⁻¹



VANADIUM(V) OXIDE CATALYST Since the forward reaction is exothermic, low temperature would result in a high yield of $SO_3(g)$ · at 1 atm pressure the yield is already very high at 97% · conditions used: temperature 450 °C, pressure 2 atm.

FIG — *The yield is already very high — about 97% at 450 °C and 1 atm pressure.*

The mechanism of action of the vanadium(V) oxide is different from iron in the Haber process. The vanadium changes its oxidation state during the reaction but converts back to its original oxidation state. The mechanism for the reaction is

$$SO_2(g) + V_2O_5(s) \rightleftharpoons SO_3(g) + V_2O_4(s)$$

$$V_2O_4(s) + \tfrac{1}{2}O_2(g) \rightleftharpoons V_2O_5(s)$$

Conditions used in the Contact process are;

TEMPERATURE	PRESSURE
450 °C	2 atm

END OF TOPIC 09

01 · Formulae, Equations and Amount of Substance
02 · Atomic Structure and the Periodic Table
03 · Bonding and Structure
04 · Introductory Organic Chemistry and Alkanes
05 · Alkenes
06 · Energetics
07 · Intermolecular Forces
08 · Redox Chemistry and Groups 1, 2 and 7
10 · Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

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